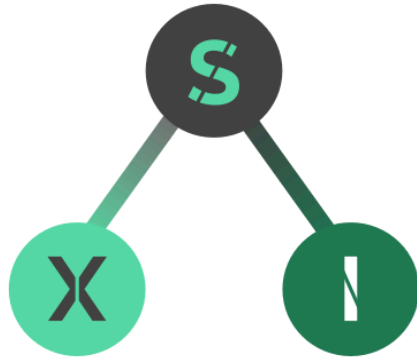




XBee Sniffer Interface - Help

Welcome to the XSI help documentation. This document contains information about the application to aid you in times of troubleshooting or understanding the application's features. If you're reading this, this is the official documentation for the app, which means you likely already have installed it from the Github repository. Check out the repository for more information on installations, versions, and releases.

None

IMPORTANT

The current release available on the Github repository is Version 0.3.0 with executable and python versions of the application.



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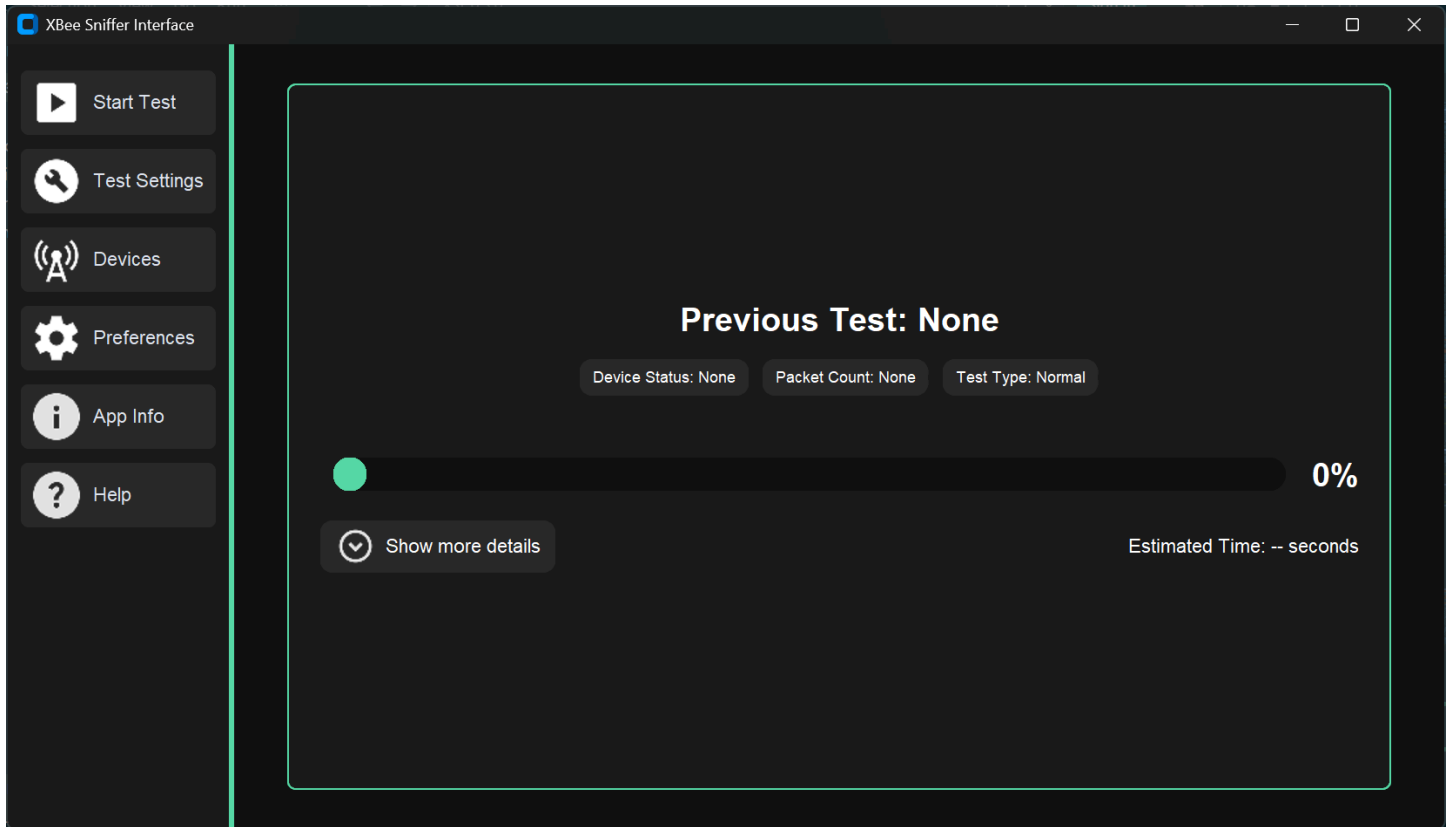
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Application Startup

Upon startup, the application should open up like shown below. This is the main window for the application. It features a navigation bar on the left with a main testing frame on the middle / right.

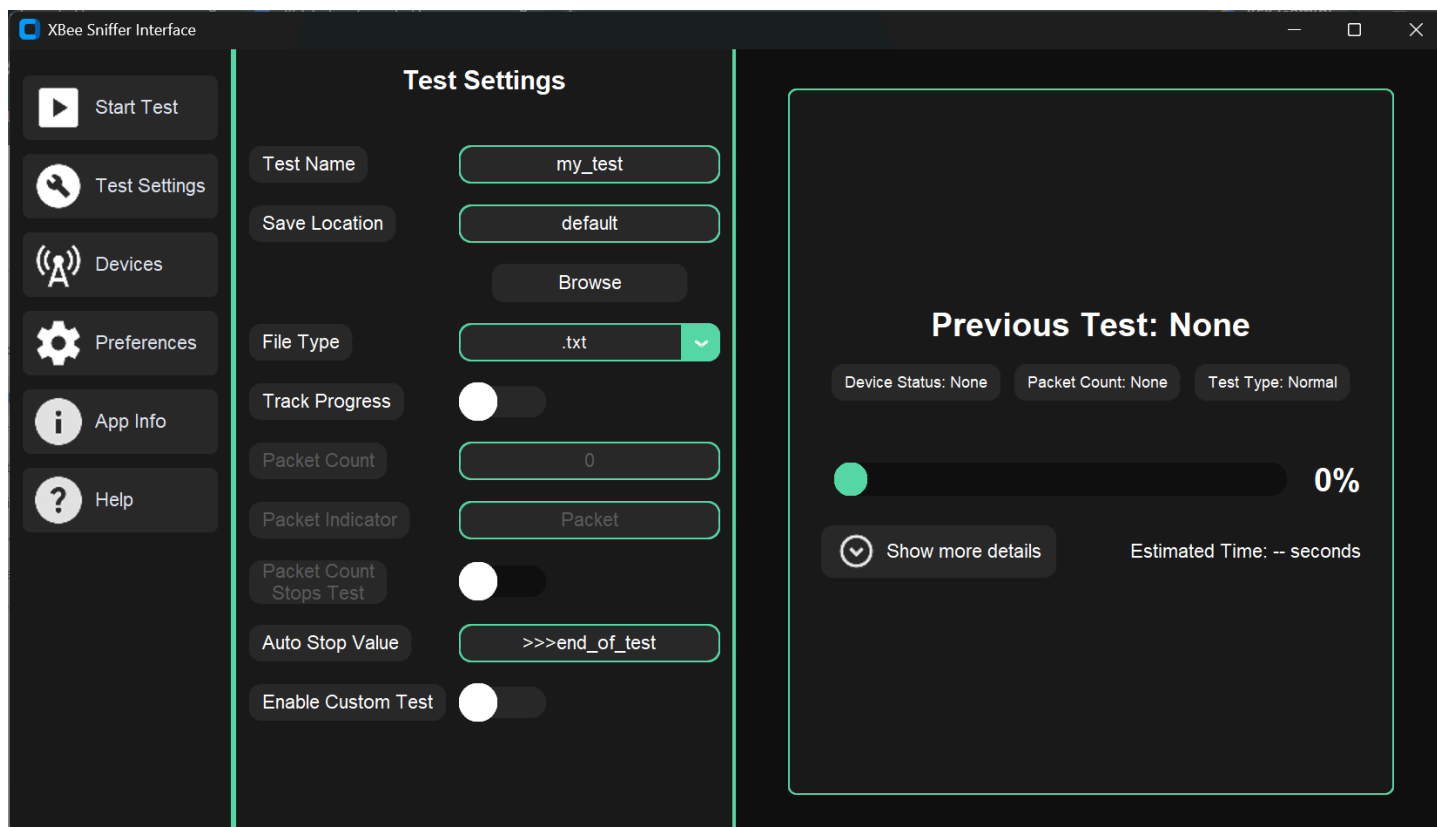


All actions can be completed through the navigation bar on the left. When the window size decreases to a certain width, the navigation panel will also resize to be smaller to help compensate for the testing frame.



Navigation Panel

Test Settings Menu



The test settings menu can be accessed in the navigation panel right below the start test button. This is where all the default test settings can be changed and configured. Test name, file location, and file type can be easily changed. XSI will save every test once completed, either in a chosen location or the default location. The default location is located in the XSI folder when installed inside a folder named 'cache.'

None

IMPORTANT

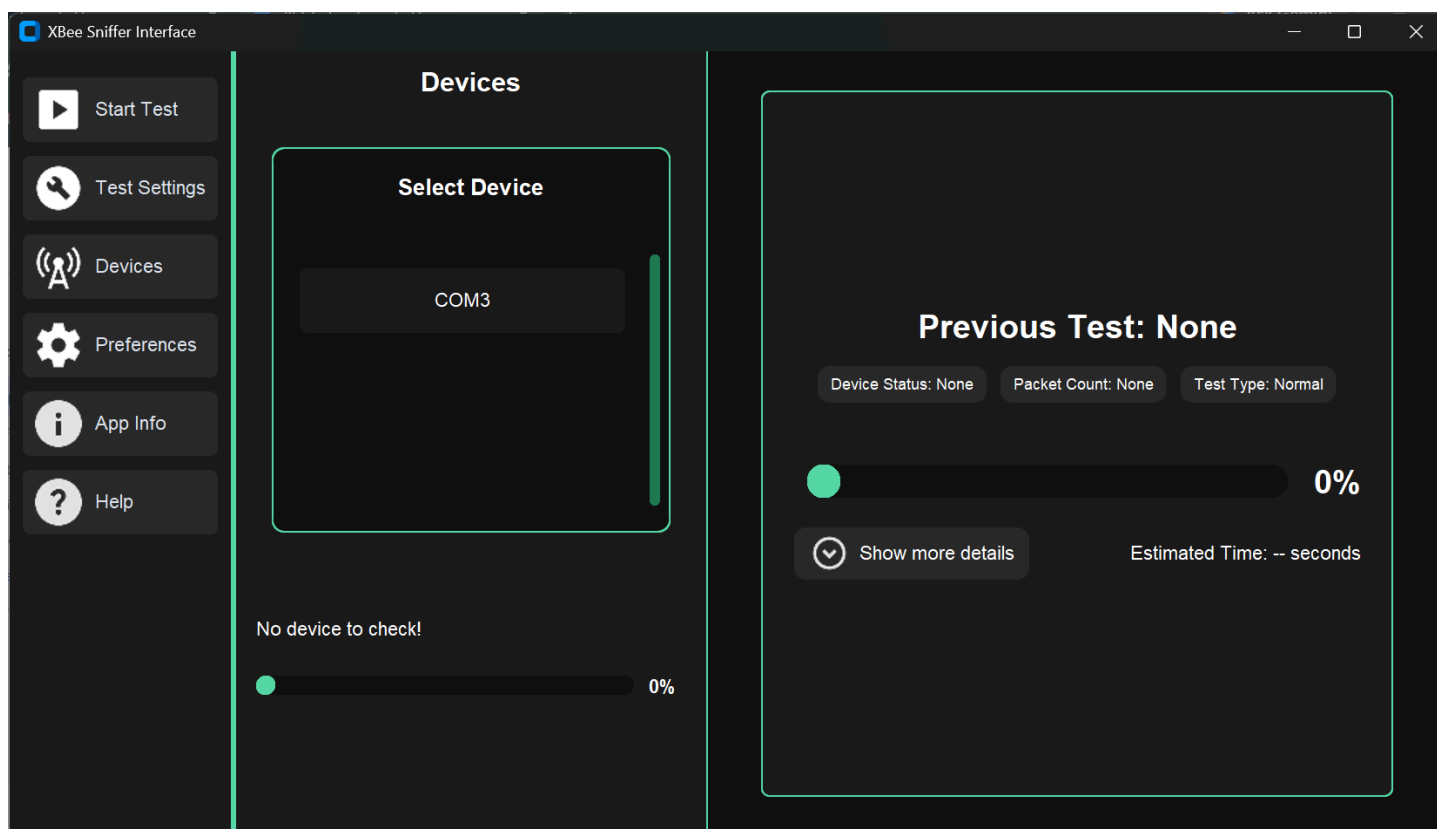
The only supported file types currently are .txt and .csv.

The next feature is tracking progress. If disabled, XSI will simply play an animation in the progress bar, and the progress number will be replaced with a dial. The dial will only spin when data is collected. If enabled, XSI has the ability to track progress throughout the test. It can even stop the test when it reaches the specified packet amount. However, if you only want the test to end from a



specified string from the network, then you don't have to enable that, and instead, use the auto stop value feature. Simply type in the value that XSI should look out for, and if it sees that, it will automatically stop the test. Another option is the packet indicator. This is how XSI can recognize what is a packet. It simply looks at the data and tries to find that value in it. The last feature is a new one called 'Enable Custom Test.' This will be explored later in the **Custom Testing** section.

Device Menu



The device menu can be accessed using the navigation panel and is found under the 'Test Settings' button. The device menu allows the user to connect to one of their 'XBee sniffer' devices. Choosing and ensuring that the correct device has been selected is essential. When you click on any device in the 'Select Device' scrollable frame, XSI will immediately start to do a check to verify that the device is an XBee sniffer. The text at the bottom can have two different responses. The first one being 'Device Response: FAILED' which means that the device didn't respond in time or is not an XBee sniffer. The other response is 'Device Response: OK!' which means that the device has been confirmed to be an XBee sniffer. After this, this is the device that XSI will use, so it is automatically stored in the software, in which case, you can exit the device frame safely.



None

WARNING

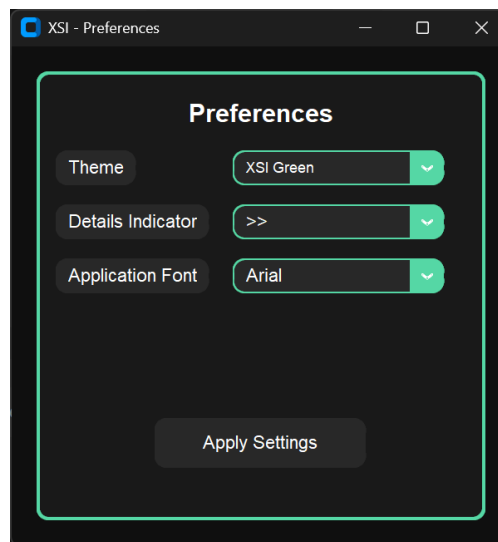
Avoid closing the device frame if it is currently checking the device. While this is not necessarily harmful, (as the loop still runs in the background) you may actually not have a real device and won't be able to tell.

None

IMPORTANT

If the device is not showing up when your in the device frame, and you've just recently plugged something in, please re-open the device frame as this will cause it to refresh.

Preferences Window



The preferences window can be accessed using the navigation panel and is under the 'Devices' button. The preferences window allows you to change small aspects of the app's overall appearance. Three options are the theme color, details indicator, and the font used in the application. There are 6 different themes to choose from, all of which change the color of the application. The details indicator is for the testing frame when showing information flowing in the terminal listbox. There are 4 different fonts to choose from as well.

None

IMPORTANT

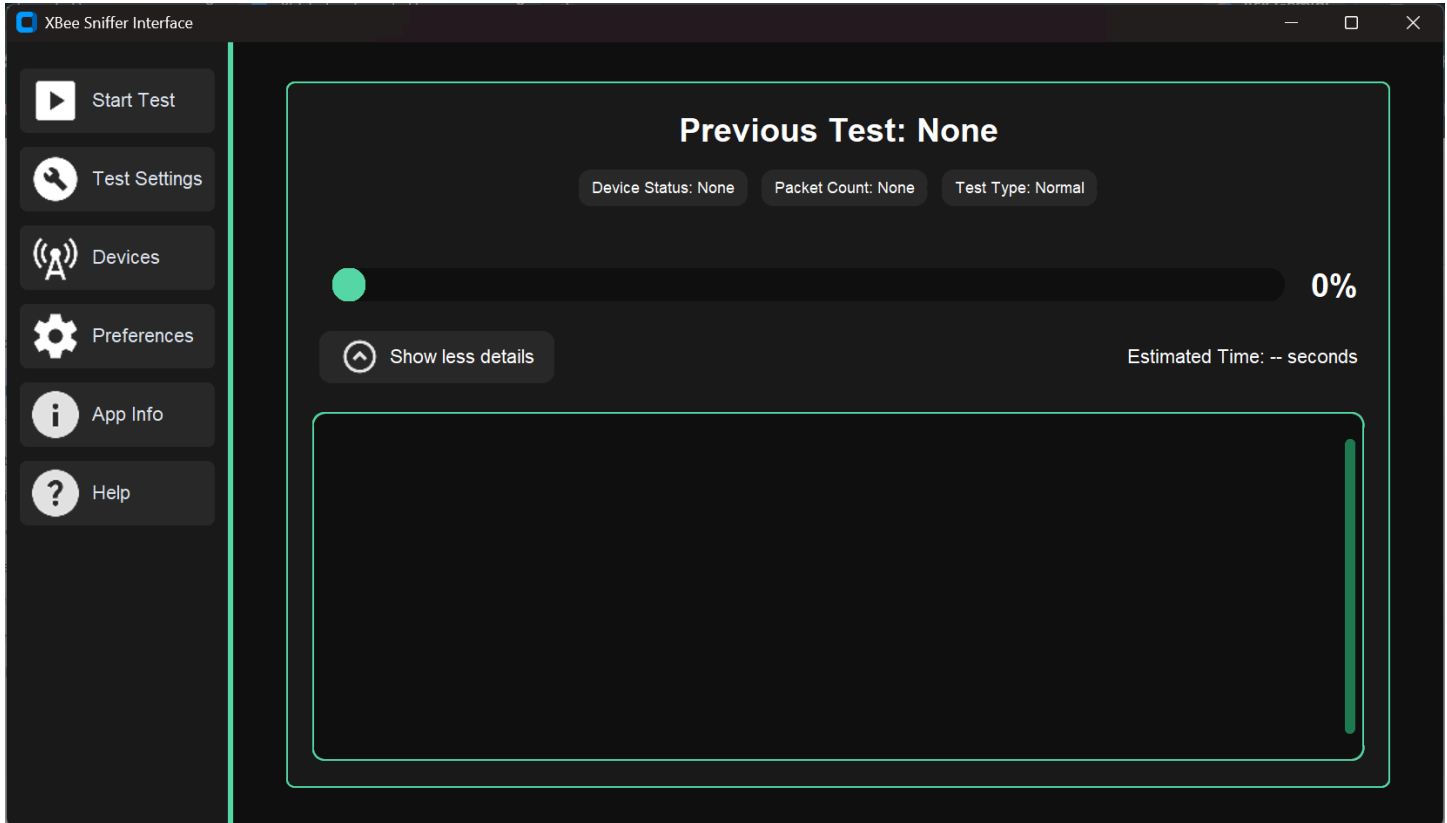
These themes are currently not saved upon closing the application as of Version 0.3.0.

Info and Help

The application info can be accessed from the navigation panel and will show a frame giving some important information on the application. This includes app name, version number, UI framework, license agreement, and the operating system. The help button will take you to this PDF in the default application for viewing and reading PDF files.



Testing Panel



Starting Test

Once you've gotten all of the test settings configured and the device setup, you can click on the 'Start Test' button in the navigation panel. Once done, 'Test Settings' and 'Devices' will be disabled until the test is stopped (either manually or automatically).

Once you have the test running, the testing frame will update automatically with all the available info that you've inputted into it. In the top, the 'Test Running: [TEST_NAME]' will display to show what test is currently running. Below are three other status texts. The 'Device Status: [STATUS]' will display whether the device actually connected or not during the test. 'Packet Count: [AMOUNT]' will show what the packet amount was set to (if track progress is turned off, it will read 0). And lastly, 'Test Type: [TEST_TYPE]' will show whether you enabled a custom test or not. The progress bar and text will update as the test goes on. If the tracking progress is not enabled, then it will simply animate. For more information, please read the sub-section **Test Settings Menu** in the section **Navigation Panel**.

Below the progress bar is the button 'Show more details' and a time estimator. The time estimator will display with what the program thinks the remaining time left in the test based on how fast the packets are coming in. The details listbox is enabled through the 'Show more details' button below the

progress bar. This will display the current data incoming through the stream so that you can view it in real time.

Stopping Test

You can stop the test any time you want to manually if chosen to do so. You'll notice that when the test starts running the 'Start Test' button will turn into a 'Stop Test' button. This will show a pop-up notification asking for confirmation about your choice. Once done, the test will end and save a file output of it in the desired location. Otherwise, if you've enabled one of the two different auto stop features, the test will end on its own when it's complete, and save a file output of what it collected.

None

CAUTION

Be careful about simply closing the program while a test is running. This does not have a notifying warning and will exit the program immediately!

File Output

At the end of every test, a file output is created of the data collected by the software. The file output has a header named 'Test Parameters' and contains info on the following: file name, packet count, XBee port name, time started, and the date conducted of the test. The rest of the file should be the data collected. Depending on the settings chosen, it could be read as either a text file (txt) or as comma separated values (csv).

None

WARNING

If you name a file with the same name as the previous test you started, XSI will simply overwrite and delete that originally file when a new test is started!



Custom Testing

XSI offers custom testing and allows the user to build out their own test. In order to do this, you need to navigate to the XSI folder > bin > tools. In the tools folder, a python file called 'xsi_custom_test.py' will be in there. It's imperative that you do not change the name since this is how XSI finds the custom test file. If you open the python file, you'll find that some of it is filled out, and a template has already been made within the file. The python file does contain comments to hint how to create your own custom test to run. This will only run if you enable the custom test option in the test settings menu.

None

IMPORTANT

This document does not provide a tutorial on Python programming or how to program with the UI framework being used. To find out more about the UI framework, click [here](#).

None

CAUTION

The custom test python file is not guaranteed to work with the application, it is advised you code and use at your own risk, and code with caution. Deleting anything that was actually supposed to be there may break the application!